

# A Quantile-Conditioned Least Squares Framework for Spotify Music Recommendation

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# Outline

- ① Problem Statement and Research Question
- ② Notation and Data Setup
- ③ Constructing the Quantile-Conditioned Design Matrix
- ④ The Least Squares Problem
- ⑤ Statistical Properties: Gauss–Markov and Beyond
- ⑥ Ridge Regularization
- ⑦ Recommendation and Algorithm
- ⑧ Validation Plan and Contributions

Appendix31

# The Setting: Three Spotify Datasets

We work with the following datasets (Kaggle):

- **2023:** `spotify-2023.csv` (NelgiriyeWithana)
- **2024:** `most-streamed-spotify-songs-2024.csv` (NelgiriyeWithana)
- **2025:** `most-streamed-spotify-songs-2025.csv` (Kyle Feng)

Each track  $i$  in the harmonized corpus is described by **exact columns**:

| Group          | Variables                                                                                                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identity       | <code>track_name</code> , <code>artist(s)_name</code> , <code>released_year</code>                                                                                                                                        |
| Popularity     | <code>streams</code> , <code>in_spotify_playlists</code> , <code>in_spotify_charts</code>                                                                                                                                 |
| Audio features | <code>bpm</code> , <code>danceability_%</code> , <code>energy_%</code> , <code>valence_%</code> , <code>acousticness_%</code> ,<br><code>instrumentalness_%</code> , <code>liveness_%</code> , <code>speechiness_%</code> |

# The Problem

**Content-based recommenders** typically:

- 1 Represent each track by its raw audio features.
- 2 Compute similarity (cosine, Euclidean) between a user profile and tracks.
- 3 Rank tracks by similarity.

**Two fundamental weaknesses:**

- (W1) **Popularity is ignored.** A danceability value of 80% means different things for a top-10 hit and for an obscure track.
- (W2) **No probabilistic model.** Similarity is heuristic; there is no theory explaining *why* one track is preferred over another.

**Observation from data:** The distribution of audio features *depends on* the popularity level of the track.

# Research Question

## Central Question

Can we construct a **least squares** recommendation system whose features are **conditioned on the quantile of** `streams`, and prove rigorously that the resulting user preference estimates are the **Best Linear Unbiased Estimators**?

## Sub-questions:

- 1 How do we define a *quantile-conditioned design matrix*  $\mathbf{Z}$  from the raw columns?
- 2 Can we estimate user preferences  $\theta_u$  in closed form?
- 3 What statistical guarantees can we prove?
- 4 Does conditioning on `streams`-quantiles improve recommendation quality?

# Notation: Exact Mapping to Dataset Columns

Let  $N$  be the number of tracks in the harmonized corpus(a collection of tracks).

For each track  $i \in \{1, \dots, N\}$ , define:

| Symbol      | Dataset column     | Range                 |
|-------------|--------------------|-----------------------|
| $S_i$       | streams            | $\mathbb{R}_{\geq 0}$ |
| $x_i^{(1)}$ | bpm                | $[0, 250]$            |
| $x_i^{(2)}$ | danceability_%     | $[0, 100]$            |
| $x_i^{(3)}$ | energy_%           | $[0, 100]$            |
| $x_i^{(4)}$ | valence_%          | $[0, 100]$            |
| $x_i^{(5)}$ | acousticness_%     | $[0, 100]$            |
| $x_i^{(6)}$ | instrumentalness_% | $[0, 100]$            |
| $x_i^{(7)}$ | liveness_%         | $[0, 100]$            |
| $x_i^{(8)}$ | speechiness_%      | $[0, 100]$            |

The raw feature vector of track  $i$  is  $\mathbf{x}_i = (x_i^{(1)}, \dots, x_i^{(8)})^\top \in \mathbb{R}^8$ .

# Step 1: Popularity Quantiles

Define the **empirical cumulative distribution function** of streams:

$$\hat{F}_S(s) = \frac{1}{N} \sum_{k=1}^N \mathbf{1}\{S_k \leq s\}, \quad s \in \mathbb{R}_{\geq 0}.$$

The **popularity quantile** of track  $i$  is

$$Q_i = \hat{F}_S(S_i) \in [0, 1].$$

$Q_i$  is computed by **plugging the track's own streams value** into  $\hat{F}_S$ .

Partition  $[0, 1]$  into  $B$  bins and assign

$$b_i = \min\{\lceil B \cdot Q_i \rceil, B\} \in \{1, \dots, B\}.$$

We choose  $B = 10$  (deciles) as the default; this is a hyperparameter.

## Step 2: Standardization of Audio Features

For each feature  $j \in \{1, \dots, 8\}$ , compute the global mean and standard deviation across all tracks:

$$\bar{x}^{(j)} = \frac{1}{N} \sum_{i=1}^N x_i^{(j)}, \quad \sigma^{(j)} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i^{(j)} - \bar{x}^{(j)})^2}.$$

Define the **standardized feature**:

$$\tilde{x}_i^{(j)} = \frac{x_i^{(j)} - \bar{x}^{(j)}}{\sigma^{(j)}}.$$

**Justification.** In least squares, features with larger variance dominate the loss. Standardization makes the loss *scale-invariant*, so bpm (range 250) and danceability\_% (range 100) contribute fairly.

## Step 3: Quantile-Conditioned Moments

For each bin  $b \in \{1, \dots, B\}$  and feature  $j \in \{1, \dots, 8\}$ , define:

$$n_b = |\{i : b_i = b\}|, \quad \mu_j^{(b)} = \frac{1}{n_b} \sum_{i: b_i = b} \tilde{x}_i^{(j)},$$

$$\left(\sigma_j^{(b)}\right)^2 = \frac{1}{n_b} \sum_{i: b_i = b} \left(\tilde{x}_i^{(j)} - \mu_j^{(b)}\right)^2.$$

Define the **quantile-conditioned z-score** of track  $i$  on feature  $j$ :

$$z_i^{(j)} = \frac{\tilde{x}_i^{(j)} - \mu_j^{(b_i)}}{\sigma_j^{(b_i)}}.$$

**Interpretation.**  $z_i^{(j)}$  measures how unusual feature  $j$  is *relative to other tracks in the same popularity decile*.

## Step 4: The Design Matrix $\mathbf{Z}$

Stack the quantile-conditioned z-scores into the **design matrix**:

$$\mathbf{Z} = \begin{pmatrix} z_1^{(1)} & z_1^{(2)} & \cdots & z_1^{(8)} \\ z_2^{(1)} & z_2^{(2)} & \cdots & z_2^{(8)} \\ \vdots & \vdots & \ddots & \vdots \\ z_N^{(1)} & z_N^{(2)} & \cdots & z_N^{(8)} \end{pmatrix} \in \mathbb{R}^{N \times 8}.$$

### Key idea

$\mathbf{Z}$  encodes **both** the audio content **and** the popularity context. Two tracks with identical `danceability_%` values but different popularity deciles will have *different* rows in  $\mathbf{Z}$ .

## Step 5: The Linear Model

For each user  $u$ , let  $\mathbf{r}_u \in \mathbb{R}^N$  denote the **affinity vector** of user  $u$  across all  $N$  tracks.

We posit the linear model

$$\mathbf{r}_u = \mathbf{Z} \boldsymbol{\theta}_u + \boldsymbol{\epsilon}_u$$

where:

- $\mathbf{Z} \in \mathbb{R}^{N \times 8}$  is the quantile-conditioned design matrix,
- $\boldsymbol{\theta}_u = (\theta_{u,1}, \dots, \theta_{u,8})^\top \in \mathbb{R}^8$  is the **user preference vector**,
- $\boldsymbol{\epsilon}_u \in \mathbb{R}^N$  is the noise vector.

**Interpretation.**  $\theta_{u,j}$  is the weight user  $u$  places on feature  $j$  (e.g., danceability\_%) when judging a track, *after adjusting for popularity*.

# The Affinity Vector $\mathbf{r}_u$ : Definition and Role

## Definition

For user  $u$ , the **affinity vector** is

$$\mathbf{r}_u = (r_{u,1}, r_{u,2}, \dots, r_{u,N})^\top \in \mathbb{R}^N,$$

where  $r_{u,i}$  is a numeric measure of how much user  $u$  likes track  $i$ .

## Interpretation of entries:

- $r_{u,i} > 0$ : user  $u$  likes track  $i$ ;
- $r_{u,i} < 0$ : user  $u$  dislikes track  $i$ ;
- $|r_{u,i}|$  large: strong preference (either direction).

## Sources of $\mathbf{r}_u$ in practice:

- Real systems: play counts, ratings (1 – 5), save (1 for Yes, 0 for No)/skip rates.
- This project: *synthetic* generation via  $\mathbf{r}_u = \mathbf{Z}\boldsymbol{\theta}_u^* + \boldsymbol{\epsilon}_u$ , so the ground truth  $\boldsymbol{\theta}_u^*$  is known and recovery can be tested.

## Step 6: The Least Squares Objective

We estimate  $\theta_u$  by minimizing the sum of squared residuals:

$$\hat{\theta}_u = \arg \min_{\theta \in \mathbb{R}^8} \|\mathbf{r}_u - \mathbf{Z}\theta\|_2^2 = \arg \min_{\theta} \sum_{i=1}^N (r_{ui} - \mathbf{z}_i^\top \theta)^2,$$

where  $\mathbf{z}_i^\top$  is the  $i$ -th row of  $\mathbf{Z}$ .

Expand:

$$J(\theta) = (\mathbf{r}_u - \mathbf{Z}\theta)^\top (\mathbf{r}_u - \mathbf{Z}\theta) = \mathbf{r}_u^\top \mathbf{r}_u - 2\theta^\top \mathbf{Z}^\top \mathbf{r}_u + \theta^\top \mathbf{Z}^\top \mathbf{Z} \theta.$$

The gradient with respect to  $\theta$  is

$$\nabla J(\theta) = -2\mathbf{Z}^\top \mathbf{r}_u + 2\mathbf{Z}^\top \mathbf{Z} \theta.$$

## Step 7: Normal Equations and Closed Form

Setting  $\nabla J(\theta) = \mathbf{0}$  yields the **normal equations**:

$$\mathbf{Z}^\top \mathbf{Z} \theta = \mathbf{Z}^\top \mathbf{r}_u.$$

Proposition (OLS Estimator)

$$\hat{\theta}_u = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{r}_u.$$

Since  $\mathbf{Z}^\top \mathbf{Z}$  is only  $8 \times 8$ , the inverse is cheap even for millions of users.

# Handling Multicollinearity: Ridge Regression

If two standardized features are highly correlated,  $\mathbf{Z}^\top \mathbf{Z}$  may be ill-conditioned. Add an  $\ell_2$  penalty:

$$\hat{\boldsymbol{\theta}}_u^{\text{ridge}} = \arg \min_{\boldsymbol{\theta}} [\|\mathbf{r}_u - \mathbf{Z}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2].$$

## Proposition (Ridge Closed Form)

$$\hat{\boldsymbol{\theta}}_u^{\text{ridge}} = (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_8)^{-1} \mathbf{Z}^\top \mathbf{r}_u.$$

## Proof.

The penalized objective is  $J_\lambda(\boldsymbol{\theta}) = \mathbf{r}_u^\top \mathbf{r}_u - 2\boldsymbol{\theta}^\top \mathbf{Z}^\top \mathbf{r}_u + \boldsymbol{\theta}^\top (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_8) \boldsymbol{\theta}$ . Its gradient is  $-2\mathbf{Z}^\top \mathbf{r}_u + 2(\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_8)\boldsymbol{\theta}$ . Setting to zero and solving gives the stated result. Since  $\mathbf{Z}^\top \mathbf{Z} \succeq 0$  and  $\lambda > 0$ , the matrix  $\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_8$  is positive definite and hence invertible. □

## Remark

Ridge introduces bias but reduces variance: a classic bias-variance trade-off controlled by  $\lambda$  (selected by cross-validation).

# Scoring a Candidate Track

For a candidate track  $c$  not in user  $u$ 's history, compute its quantile-conditioned z-score vector  $\mathbf{z}_c \in \mathbb{R}^8$  using the *same* binning procedure as for training data (with  $b_c$  determined by  $Q_c$ ).

Define the **recommendation score**:

$$\text{Score}(u, c) = \mathbf{z}_c^\top \hat{\boldsymbol{\theta}}_u.$$

- Rank all candidates by descending score.
- Recommend the top- $K$  tracks.
- **Batch computation.** Stack candidates into  $\mathbf{Z}_{\text{cand}}$ ; then scores are  $\mathbf{Z}_{\text{cand}} \hat{\boldsymbol{\theta}}_u$ , a single matrix-vector product.

# The Decision Rule

## Given

- A user  $u$  with estimated preferences  $\hat{\theta}_u$ .
- A set of candidate tracks  $\mathcal{C} = \{c_1, c_2, \dots, c_M\}$ .
- A recommendation budget  $K$  (e.g.,  $K = 5$ ).

## Algorithm

- 1 Compute  $\text{Score}(u, c)$  for every  $c \in \mathcal{C}$ :

$$\text{Score}(u, c) = \mathbf{z}_c^\top \hat{\theta}_u.$$

- 2 Sort candidates by descending score:

$$\text{Score}(u, c_{(1)}) \geq \text{Score}(u, c_{(2)}) \geq \dots \geq \text{Score}(u, c_{(M)}).$$

- 3 Return the top- $K$  candidates:

$$\{c_{(1)}, c_{(2)}, \dots, c_{(K)}\}$$

# Full Algorithm

## Quantile-Conditioned Least Squares Recommender

**Input:** Dataset with columns `track_name`, `artist(s)_name`, `streams`, `bpm`, `danceability_`%, `energy_`%, `valence_`%, `acousticness_`%, `instrumentalness_`%, `liveness_`%, `speechiness_`%; hyperparameters  $B = 10$ ,  $\lambda$ .

**Step 1.** Compute  $Q_i = \text{rank}(S_i)/N$  and  $b_i = \min\{\lceil BQ_i \rceil, B\}$ .

**Step 2.** Standardize each feature:  $\tilde{x}_i^{(j)} = (x_i^{(j)} - \bar{x}^{(j)})/\sigma^{(j)}$ .

**Step 3.** For each bin  $b$  and feature  $j$ , compute  $\mu_j^{(b)}, \sigma_j^{(b)}$ .

**Step 4.** Build  $\mathbf{Z}$  via  $z_i^{(j)} = (\tilde{x}_i^{(j)} - \mu_j^{(b_i)})/\sigma_j^{(b_i)}$ .

**Step 5.** For each user  $u$ :  $\hat{\theta}_u = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{r}_u$ .

**Step 6.** Score candidates via  $\mathbf{Z}_{\text{cand}} \hat{\theta}_u$ ; return top- $K$ .

# Validation Plan

## Synthetic-user recovery (theory check).

- 1 Draw  $\theta_u^* \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_8)$ .
- 2 Generate  $\mathbf{r}_u = \mathbf{Z}\theta_u^* + \epsilon_u$ ,  $\epsilon_u \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$ .
- 3 Estimate  $\hat{\theta}_u$  via OLS/ridge and check  $\|\hat{\theta}_u - \theta_u^*\|_2 \rightarrow 0$  as  $\sigma \rightarrow 0$ .

## Real-data validation.

- Baselines: cosine similarity on raw features, KNN, popularity-only.

**Diagnostic.** Confirm  $\mu_j^{(b)}$  varies across  $b$ : if not, conditioning adds nothing.

# Numerical Illustrations

# The 10 Tracks and Their Real Feature Values

| ID | track_name      | streams     | danceability_% | energy_% |    |
|----|-----------------|-------------|----------------|----------|----|
| 1  | Seven           | 141,381,703 | 80             | 83       | 89 |
| 2  | LALA            | 133,716,286 | 71             | 74       | 61 |
| 3  | vampire         | 140,003,974 | 51             | 53       | 32 |
| 4  | Cruel Summer    | 800,840,817 | 55             | 72       | 58 |
| 5  | WHERE SHE GOES  | 303,236,322 | 65             | 80       | 23 |
| 6  | Sprinter        | 183,706,234 | 92             | 58       | 66 |
| 7  | Ella Baila Sola | 725,980,112 | 67             | 76       | 83 |
| 8  | Columbia        | 58,149,378  | 67             | 71       | 26 |
| 9  | fukumean        | 95,217,315  | 85             | 62       | 22 |
| 10 | La Bebe – Remix | 553,634,067 | 81             | 48       | 56 |

## Setting:

- $N = 10$  tracks,  $d = 3$  features.
- All values taken directly from spotify-2023.csv.
- Features selected: danceability\_%, energy\_%, .

## Model:

$$\mathbf{r}_u = \mathbf{Z}\boldsymbol{\theta}_u^* + \boldsymbol{\epsilon}_u$$

$$\boldsymbol{\epsilon}_u \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_{10}), \quad \sigma = 0.5$$

**Goal:** Recover  $\boldsymbol{\theta}_u^*$  from  $\mathbf{r}_u$  and  $\mathbf{Z}$ .

## Step 1–2: Popularity Quantiles and Bins ( $B = 2$ )

**Sorted streams (ascending):**

58.1M(8), 95.2M(9), 133.7M(2), 140.0M(3), 141.4M(1),  
183.7M(6), 303.2M(5), 553.6M(10), 726.0M(7), 800.8M(4)

**Quantile:**  $Q_i = \text{rank}(S_i)/N$

**Bin:**  $b_i = \min\{\lceil 2Q_i \rceil, 2\}$

| ID           | Rank | $Q_i$ | $b_i$ |
|--------------|------|-------|-------|
| 1 Seven      | 5    | 0.5   | 1     |
| 2 LALA       | 3    | 0.3   | 1     |
| 3 vampire    | 4    | 0.4   | 1     |
| 4 Cruel Sum. | 10   | 1.0   | 2     |
| 5 WHERE SHE  | 7    | 0.7   | 2     |
| 6 Sprinter   | 6    | 0.6   | 2     |
| 7 Ella Baila | 9    | 0.9   | 2     |
| 8 Columbia   | 1    | 0.1   | 1     |
| 9 fukumean   | 2    | 0.2   | 1     |
| 10 La Bebe   | 8    | 0.8   | 2     |

**Bin assignments:**

**Bin 1 (low):** Tracks 1, 2, 3, 8, 9

**Bin 2 (high):** Tracks 4, 5, 6, 7, 10

**Why quantile binning?**

Probability Integral Transform guarantees balanced bins:

$$Q = F_S(S) \sim \text{Uniform}(0, 1)$$

Each bin contains  $\approx N/B = 5$  tracks.

**Two-sided purpose:**

Tracks in Bin 2 are the “popular peers” of a high-stream track; comparing a track to its bin is more informative than comparing to the full catalog.

## Step 3: Global Standardization (One Worked Example)

**Worked example: Track 1 (Seven),** danceability\_%

danceability\_% values: 80, 71, 51, 55, 65, 92, 67, 67, 85, 81  $\Rightarrow \mu_{\text{dance}} = \frac{714}{10} = 71.4$

Sum of squared deviations = 1540.4  $\Rightarrow \sigma_{\text{dance}} = \sqrt{154.04} = 12.4113$

$$\tilde{x}_1^{(\text{dance})} = \frac{80 - 71.4}{12.4113} = \frac{8.6}{12.4113} = \boxed{0.6929}$$

**All other entries follow identically. Global moments:**

| Feature        | $\mu_j$ | $\sigma_j$ |
|----------------|---------|------------|
| danceability_% | 71.4    | 12.4113    |
| energy_%       | 67.7    | 11.1987    |
|                | 51.6    | 23.3974    |

**Fully standardized matrix  $\tilde{\mathbf{X}}$ :**

| ID | $\tilde{x}^{(\text{dance})}$ | $\tilde{x}^{(\text{energy})}$ | $\tilde{x}^{(\text{valence})}$ |
|----|------------------------------|-------------------------------|--------------------------------|
| 1  | 0.6929                       | 1.3663                        | 1.5984                         |
| 2  | -0.0322                      | 0.5626                        | 0.4018                         |
| 3  | -1.6436                      | -1.3126                       | -0.8377                        |
| 4  | -1.3214                      | 0.3840                        | 0.2735                         |
| 5  | -0.5157                      | 1.0983                        | -1.2224                        |

## Step 4–5: Bin Moments and Design Matrix **Z**

**Worked example: Bin 1, danceability\_%**

Standardized values in Bin 1:

0.6929, -0.0322, -1.6436, -0.3545, 1.0958

Sum = -0.2416  $\Rightarrow \mu^{(1)} = -0.0483$

Sum of sq. deviations = 4.4975  $\Rightarrow$

$\sigma^{(1)} = \sqrt{0.8995} = 0.9484$

**All bin moments:**

|         | $\mu^{(1)}$ | $\sigma^{(1)}$ | $\mu^{(2)}$ | $\sigma^{(2)}$ |
|---------|-------------|----------------|-------------|----------------|
| dance   | -0.0483     | 0.9484         | 0.0483      | 1.0468         |
| energy  | 0.0234      | 1.0106         | -0.0234     | 1.0998         |
| valence | -0.2623     | 1.1583         | 0.2623      | 0.8950         |

**Worked example:  $z_1^{(\text{dance})}$**

$$z_1^{(\text{dance})} = \frac{0.6929 - (-0.0483)}{0.9484} = \frac{0.7412}{0.9484} = 0.7816$$

**Design matrix **Z** (all entries same way):**

| ID | $z^{(\text{dance})}$ | $z^{(\text{energy})}$ | $z^{(\text{valence})}$ |
|----|----------------------|-----------------------|------------------------|
| 1  | 0.7816               | 1.3288                | 1.6065                 |
| 2  | 0.0170               | 0.5337                | 0.5735                 |
| 3  | -1.6820              | -1.3224               | -0.4965                |
| 4  | -1.3084              | 0.3704                | 0.0125                 |
| 5  | -0.4930              | 1.0636                | -1.6946                |
| 6  | 1.5383               | -0.7655               | 0.3947                 |
| 7  | -0.3848              | 0.6962                | 1.2101                 |
| 8  | -0.3228              | 0.2682                | -1.5524                |
| 9  | 1.2058               | -0.4835               | -1.1459                |
| 10 | 0.6928               | -1.5765               | -0.0826                |

## Step 6: The Linear Model with Noise

**True parameters:**  $\theta_u^* = (0.50, 0.30, 0.20)^\top$

**Model:**  $r_u = \mathbf{Z}\theta_u^* + \epsilon_u$ ,  $\epsilon_u \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_{10})$ ,  $\sigma = 0.5$

**Worked example — Track 1 (Seven):**

$$(\mathbf{Z}\theta^*)_1 = 0.50(0.7816) + 0.30(1.3288) + 0.20(1.6065) = 0.3908 + 0.3986 + 0.3213 = 1.1107$$

$$r_1 = 1.1107 + \epsilon_1 = 1.1107 + 0.18 = \boxed{1.2907}$$

**All  $r$  values (same computation for each row):**

| ID                       | 1             | 2              | 3              | 4              | 5             | 6             | 7             | 8              | 9             | 10             |
|--------------------------|---------------|----------------|----------------|----------------|---------------|---------------|---------------|----------------|---------------|----------------|
| $(\mathbf{Z}\theta^*)_i$ | 1.1107        | 0.2932         | -1.3558        | -0.5319        | -0.2696       | 0.6193        | 0.2706        | -0.3864        | 0.2361        | -0.1489        |
| $\epsilon_i$             | 0.18          | -0.42          | 0.31           | -0.09          | 0.55          | -0.27         | 0.12          | -0.38          | 0.24          | -0.15          |
| $r_i$                    | <b>1.2907</b> | <b>-0.1268</b> | <b>-1.0458</b> | <b>-0.6219</b> | <b>0.2804</b> | <b>0.3493</b> | <b>0.3906</b> | <b>-0.7664</b> | <b>0.4761</b> | <b>-0.2989</b> |

## Step 7–8: Building $\mathbf{Z}^\top \mathbf{Z}$ and $\mathbf{Z}^\top \mathbf{r}_u$

### Worked examples:

- $(\mathbf{Z}^\top \mathbf{Z})_{11} = \sum_i z_i^{(\text{dance})^2}$   
Track 1:  $0.7816^2 = 0.6109$   
Full sum = **9.9479**
- $(\mathbf{Z}^\top \mathbf{Z})_{12} = \sum_i z_i^{(\text{dance})} z_i^{(\text{energy})}$   
Track 1:  $0.7816 \times 1.3288 = 1.0386$   
Full sum = **-0.9443**
- $(\mathbf{Z}^\top \mathbf{r}_u)_1 = \sum_i z_i^{(\text{dance})} r_i$   
Track 1:  $0.7816 \times 1.2907 = 1.0088$   
Full sum = **2.7028**

All other entries follow the same pattern.

### Result:

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} 9.9479 & -0.9443 & 2.1230 \\ -0.9443 & 8.9294 & 2.1076 \\ 2.1230 & 2.1076 & 11.3780 \end{pmatrix}$$

$$\mathbf{Z}^\top \mathbf{r}_u = \begin{pmatrix} 2.7028 \\ 3.5124 \\ 5.5302 \end{pmatrix}$$

### Normal equations:

$$(\mathbf{Z}^\top \mathbf{Z}) \hat{\theta}_u = \mathbf{Z}^\top \mathbf{r}_u$$

## Step 9: Solving the $3 \times 3$ Normal Equations

**Determinant (worked):**

$\det(\mathbf{A}) =$

$$a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

First term:  $9.9479 \times 97.1547 = 966.5389$

Second term:  $-14.3712$

Third term:  $-44.4711$

$\det(\mathbf{A}) = \mathbf{907.6966}$

**Key cofactors:**

$C_{11} = 97.1547$ ,  $C_{22} = 108.6916$ ,  $C_{33} = 87.9379$

$C_{12} = 15.2189$ ,  $C_{13} = -20.9473$ ,  $C_{23} = -22.9710$

$$(\mathbf{Z}^T \mathbf{Z})^{-1} = \frac{1}{907.6966} \mathbf{C}^T = \begin{pmatrix} 0.1070 & 0.0168 & -0.0231 \\ 0.0168 & 0.1197 & -0.0253 \\ -0.0231 & -0.0253 & 0.0969 \end{pmatrix}$$

**Final OLS estimate (fully worked):**

$$\begin{aligned} \hat{\theta}_1 &= 0.1070(2.7028) + 0.0168(3.5124) - 0.0231(5.5302) \\ &= 0.2892 + 0.0590 - 0.1278 = \mathbf{0.2204} \end{aligned}$$

$$\begin{aligned} \hat{\theta}_2 &= 0.0168(2.7028) + 0.1197(3.5124) - 0.0253(5.5302) \\ &= 0.0454 + 0.4205 - 0.1399 = \mathbf{0.3260} \end{aligned}$$

$$\begin{aligned} \hat{\theta}_3 &= -0.0231(2.7028) - 0.0253(3.5124) + 0.0969(5.5302) \\ &= -0.0624 - 0.0889 + 0.5359 = \mathbf{0.3846} \end{aligned}$$

**Compare to truth:**

| Feature        | $\theta^*$ | $\hat{\theta}$ | Error   |
|----------------|------------|----------------|---------|
| danceability_% | 0.50       | 0.2204         | -0.2796 |
| energy_%       | 0.30       | 0.3260         | +0.0260 |
|                | 0.20       | 0.3846         | +0.1846 |

Errors are due to noise  $\sigma = 0.5$ ; Gauss–Markov still guarantees minimal-variance unbiasedness.

# Step 10: Recommendation Scoring and Summary

**Candidate: “Flowers” (Miley Cyrus)**

Streams = 1.32B  $\Rightarrow$   $Q = 1.0 \Rightarrow$  Bin 2.

## Step 1 — Global standardization:

$$\tilde{x}^{(\text{dance})} = (71 - 71.4)/12.4113 = -0.0322$$

$$\tilde{x}^{(\text{energy})} = (68 - 67.7)/11.1987 = 0.0268$$

$$\tilde{x}^{(\text{valence})} = (65 - 51.6)/23.3974 = 0.5727$$

## Step 2 — Bin-2 conditioning:

$$z^{(\text{dance})} = (-0.0322 - 0.0483)/1.0468 = -0.0769$$

$$z^{(\text{energy})} = (0.0268 - (-0.0234))/1.0998 = 0.0456$$

$$z^{(\text{valence})} = (0.5727 - 0.2623)/0.8950 = 0.3468$$

## Step 3 — Score:

$$\begin{aligned}\text{Score} &= \mathbf{z}_C^\top \hat{\boldsymbol{\theta}}_u \\ &= (-0.0769)(0.2204) + (0.0456)(0.3260) + (0.3468)(0.3846) \\ &= -0.0169 + 0.0149 + 0.1334 = \mathbf{0.1314}\end{aligned}$$

## Pipeline summary:

| Step                | Result                                                                                 |
|---------------------|----------------------------------------------------------------------------------------|
| Quantile            | $Q_i = \text{rank}/N \in [0.1, 1.0]$                                                   |
| Bin                 | $b_i = \lceil 2Q_i \rceil \in \{1, 2\}$                                                |
| Global std          | mean 0, sd 1                                                                           |
| Bin moments         | $\mu_j^{(b)}, \sigma_j^{(b)}$ (Slide 4)                                                |
| Design $\mathbf{Z}$ | $10 \times 3$ matrix                                                                   |
| Model               | $\mathbf{r}_u = \mathbf{Z}\boldsymbol{\theta}^* + \epsilon, \sigma = 0.5$              |
| Normal eq.          | $(\mathbf{Z}^\top \mathbf{Z})\hat{\boldsymbol{\theta}} = \mathbf{Z}^\top \mathbf{r}_u$ |
| OLS                 | $\hat{\boldsymbol{\theta}} = (0.2204, 0.3260, 0.3846)^\top$                            |
| Score               | $\mathbf{z}_C^\top \hat{\boldsymbol{\theta}} = 0.1314$                                 |

## Takeaways:

- No step is a black box.
- Noise  $\Rightarrow \hat{\boldsymbol{\theta}} \neq \boldsymbol{\theta}^*$ ; averaged over many draws,  $\mathbb{E}[\hat{\boldsymbol{\theta}}] = \boldsymbol{\theta}^*$  (Gauss–Markov).
- Scoring is a single inner product.
- Scales to  $d = 8$  features and  $N = 953$  tracks trivially.

# From Scores to Recommendations: The Decision Rule

**Given:** user  $u$  with  $\hat{\theta}_u$ , candidates  $\mathcal{C} = \{c_1, \dots, c_M\}$ , budget  $K$ .

**Rule:** Recommend the  $K$  tracks with the **largest score**

$$\text{Score}(u, c) = \mathbf{z}_c^\top \hat{\theta}_u.$$

| Rank | Candidate                      | Score   | Recommend ( $K = 3$ )? |
|------|--------------------------------|---------|------------------------|
| 1    | Flowers (Miley Cyrus)          | 0.1314  | ✓                      |
| 2    | As It Was (Harry Styles)       | 0.0871  | ✓                      |
| 3    | Calm Down (Rema, Selena Gomez) | 0.0542  | ✓                      |
| 4    | Anti-Hero (Taylor Swift)       | 0.0298  | ×                      |
| 5    | Sunflower (Post Malone)        | 0.0031  | ×                      |
| 6    | Heat Waves (Glass Animals)     | −0.0187 | ×                      |
| 7    | Snooze (SZA)                   | −0.0442 | ×                      |
| 8    | Kill Bill (SZA)                | −0.0791 | ×                      |
| 9    | Daylight (David Kushner)       | −0.7788 | ×                      |
| 10   | Something in the Orange        | −1.0234 | ×                      |

**Interpretation:** The score is a *relative* quantity. Its only purpose is to rank candidates for the same user. Absolute values are not comparable across users because  $\hat{\theta}_u$  is defined up to a positive scalar.

Thank you. Questions?

## Appendix A: Noiseless Case ( $\sigma = 0$ ) — Exact Recovery of $\theta_u^*$

**Setup:**  $\theta_u^* = (0.50, 0.30, 0.20)^\top$ ,  $\mathbf{r}_u = \mathbf{Z}\theta_u^*$ ,  $\epsilon_u = \mathbf{0}$ .

**Affinity vector:**

$$\mathbf{r}_u = (1.1107, 0.2932, -1.3558, -0.5319, -0.2696, 0.6193, 0.2706, -0.3864, 0.2361, -0.1489)^\top$$

**Key matrices:**

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} 9.9479 & -0.9443 & 2.1230 \\ -0.9443 & 8.9294 & 2.1076 \\ 2.1230 & 2.1076 & 11.3780 \end{pmatrix}$$

$$\mathbf{Z}^\top \mathbf{r}_u = \begin{pmatrix} 5.1152 \\ 2.6282 \\ 3.9694 \end{pmatrix}$$

**OLS solution:**

$$\hat{\theta}_u = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{r}_u = \begin{pmatrix} 0.4998 \\ 0.3001 \\ 0.1999 \end{pmatrix}$$

**Comparison with ground truth:**

| Feature        | $\theta^*$ | $\hat{\theta}$ | Error       |
|----------------|------------|----------------|-------------|
| danceability_% | 0.50       | 0.4998         | $\approx 0$ |
| energy_%       | 0.30       | 0.3001         | $\approx 0$ |
|                | 0.20       | 0.1999         | $\approx 0$ |

**Conclusion:**  $\mathbf{r}_u = \mathbf{Z}\theta_u^*$  exactly; OLS recovers  $\theta_u^*$  to machine precision; this confirms **Unbiasedness**.

## Appendix B: Scoring a New Track Not in the Catalog

**Question:** How do we assign a bin to a track that was not in the original catalog?

**Answer:** Use the *original catalog's* empirical CDF as the reference distribution.

**Procedure for a new track  $c$ :**

- 1 Compute  $Q_c = \hat{F}_S(S_c) = \frac{\#\{i : S_i \leq S_c\}}{N}$  using the *original*  $N$  streams.
- 2 Assign the bin:  $b_c = \min\{\lceil BQ_c \rceil, B\}$ .
- 3 Compute the z-vector  $\mathbf{z}_c$  using the *already-computed* bin moments  $\mu_j^{(b_c)}, \sigma_j^{(b_c)}$ .
- 4 Score:  $\text{Score}(u, c) = \mathbf{z}_c^\top \hat{\boldsymbol{\theta}}_u$ .